
How an SMR Works

The NUWARD-type integral PWR — structure & mechanisms

A plain-language companion to the NEXUS-1 console

What this is	A walk through the smaller, simpler cousin of the large PWR: how an <i>integral</i> pressurised water reactor folds the whole reactor into one vessel, drives its water with compact in-vessel pumps, and is backed by a pool that cools it passively — explained for a curious non-specialist.
Modelled on	The integral SMR modules in the demo's fleet, named for the NUWARD design: two 170 MWe units sharing one below-grade nuclear island and spent-fuel pool. Each reactor's core, pressuriser, control-rod drives and <i>six compact plate steam generators</i> sit inside a single vessel with <i>no external loops</i> ; the primary water is moved by compact in-vessel pumps. A 76-assembly, half-height, 17×17 core, boron-free at power.
Diagrams	Four 2-D schematics — the whole plant, the integral module, the steam generator, and the fuel & core. Simplified and <i>not to scale</i> ; no dimensions.
Build	NX1-094C79E0A366D34F
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How to read these schematics. As in the PWR guide, they show *how the parts connect and how heat moves*, not how big anything is. Shapes are stylised, nothing is to scale, and there are no dimensions. The figures describe a *NUWARD-type* integral PWR using publicly described features of that design — representative and internally consistent, not a vendor drawing.

1 The idea in one minute

A small modular reactor makes electricity exactly as a large reactor does — a controlled chain reaction heats water, that heat boils a second batch of water, and the steam spins a turbine. What changes is the *packaging*. An SMR is **small** (here 170 MWe per unit instead of a thousand-plus) and **modular**: two identical units share one building, and capacity is added a unit at a time rather than poured on-site as one enormous plant.

The family the demo uses is the **integral pressurised water reactor** of the NUWARD type — in essence a PWR shrunk and folded together. Three ideas define it. First, everything a large PWR spreads across a containment building — core, steam generators, pressuriser, control-rod drives — is packed *inside one vessel*; there are no primary loops, no hot and cold legs crossing the room. Because nothing large connects to the vessel, the worst pipe break it can suffer is tiny — the largest primary connection is only about a finger's width across. Second, the water is still driven by **pumps**, but they are compact, sealed, and sit *inside* the vessel rather than on external loops. Third, each reactor sits in its own pool of water, so that after shutdown it sheds its heat **passively** into that pool, without pumps, power, or operator action. Figure 1 shows the layout.

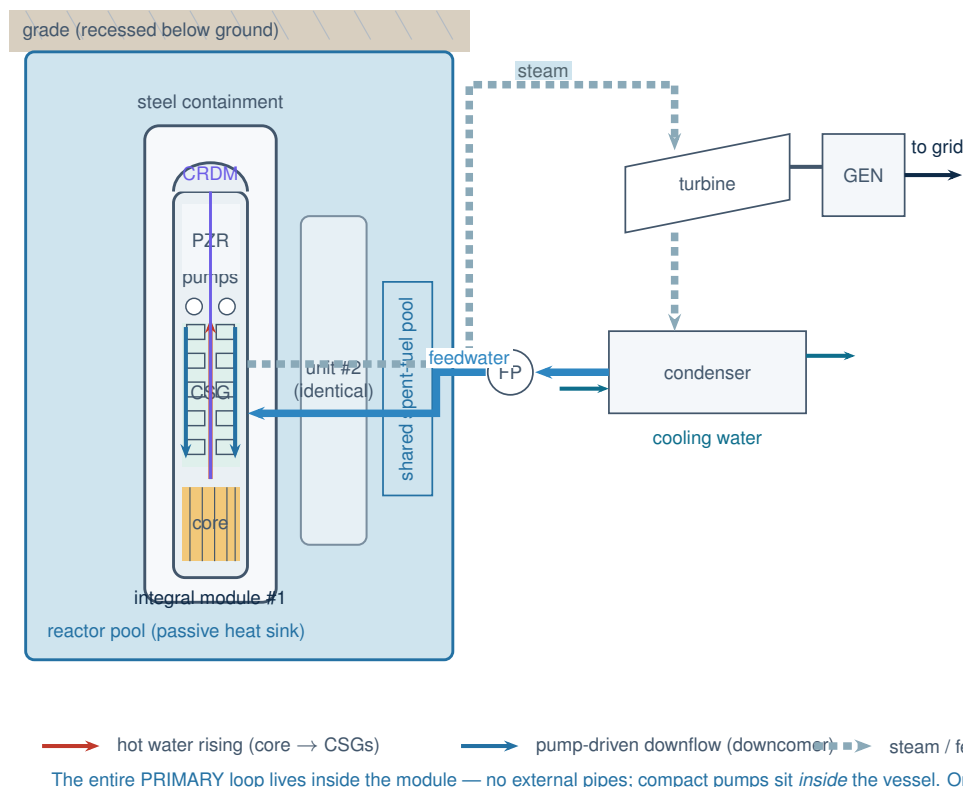


Figure 1: The whole plant. Each **integral module** holds the core, six compact plate steam generators (CSG), the in-vessel **pumps** and pressuriser (PZR) in one vessel; the water circulates inside it, pushed by the in-vessel pumps. Each module sits inside its own **steel containment**, submerged in a **pool** that serves as a passive heat sink. **Two identical 170 MWe units** share one below-grade building and a **spent-fuel pool**. Only **steam** and **feedwater** leave the vessel — to a turbine, generator and condenser, as in any plant.

Same console, smaller core. In the demo, an SMR module renders as a smaller core and joins the same fleet as the large PWR units, feeding a shared steam header — so one plant can mix big units and small modules, and the Plant 3D View shows both.

2 Following the water: the two loops

2.1 The primary loop — folded inside one vessel, driven by in-vessel pumps

The primary loop is the same idea as a large PWR's — pressurised water carrying heat away from the core — but it never leaves the vessel. Water warmed in the core rises up a central chimney (the **riser**), passes through the steam generators stacked in the outer ring (the “annulus”), and is then pushed back *down* through that ring by compact **in-vessel pumps**, returning to the bottom to flow up through the core again. Unlike some SMRs, the flow here is **pump-driven**, not gravity-driven — but because the pumps and steam generators are folded inside the vessel, there are no external loops to break, and the largest pipe connected to the vessel is only about 30 mm across, which shrinks the worst-case leak dramatically.

A **pressuriser** still keeps the loop from boiling, but in an integral reactor it is simply the steam space at the very top of the same vessel — no separate tank, no surge line. Section 3 traces the full path.

2.2 The secondary loop — clean steam to the turbine

The secondary side is essentially unchanged from a large plant, and that is the point: an SMR is unfamiliar on the inside but ordinary on the outside. Feedwater is pumped into the steam generators (which here live *inside* the reactor vessel), boils, and leaves as steam through the vessel wall. That steam crosses out of containment, spins the **turbine** and **generator**, is condensed back to water in the **condenser**, and is returned by the **feed pump**. As always, this water never touches the core's water and is not radioactive.

3 Inside the reactor module

This is where an integral SMR looks most different from a PWR: everything is stacked into one tall, slender vessel. Figure 2 cuts it open.

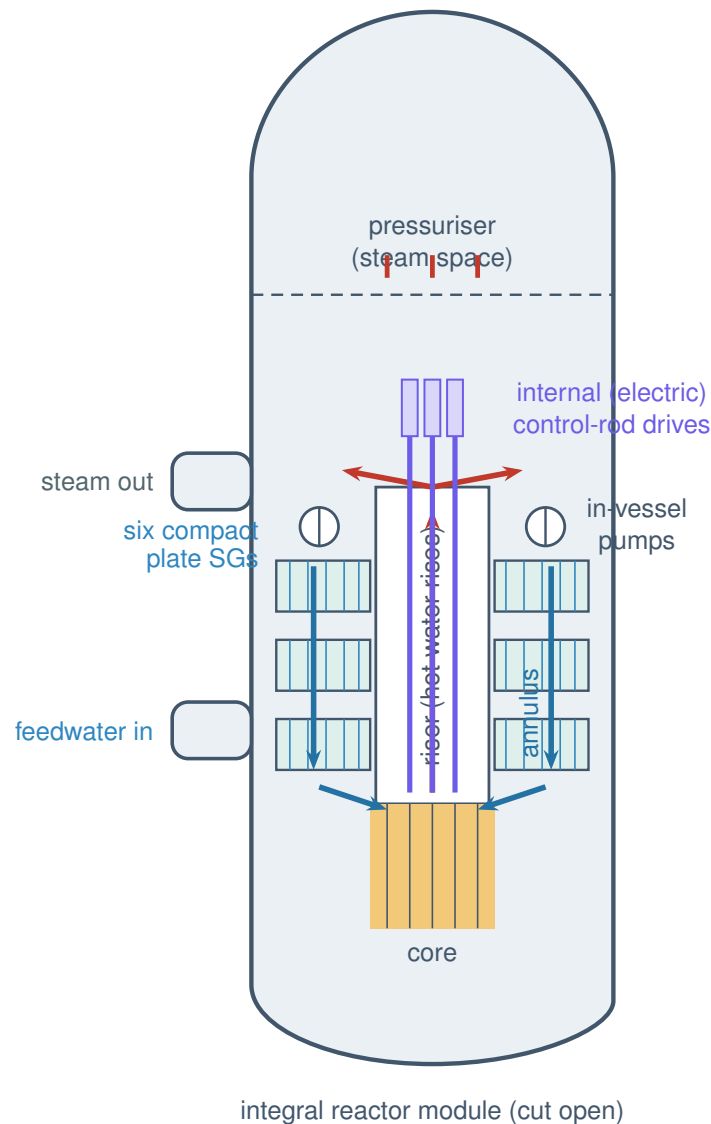


Figure 2: The integral module, cut open. Water heated in the **core** rises up the central **riser**, passes the **compact plate steam generators** in the outer **annulus**, and is pushed back down by the **in-vessel pumps** — a complete loop with no piping outside the vessel. The **pressuriser** is the steam space in the top dome, and the **control-rod drives** are electric units mounted *inside* the vessel. Only **feedwater** and **steam** cross the wall.

3.1 How the power is steered — and stopped

Reactivity is controlled mainly by **control rods** dropping into the core from above. A notable difference from a large PWR: this reactor runs **boron-free** during power operation — there is no soluble boron dissolved in the coolant for slow adjustment. Instead it leans on a strong, constant negative feedback from the water itself (the more it heats, the less

it sustains the reaction), which makes the core naturally steady. **SCRAM** is identical in spirit to any reactor — release the rods and let them drop in seconds — and the same delayed-neutron physics gives the gentle power tail the demo's engine reproduces.

What is different is what happens *after* shutdown. A reactor keeps making “decay heat” for a long time after the chain reaction stops. Here it is carried away **passively**: heat moves from the vessel into the surrounding **pool**, which can absorb it for days with no pump, power supply, or operator action. Combined with the tiny maximum pipe-break size, the design aims for a safe state that is the *default* rather than something that must be actively maintained.

4 The steam generator — where the two worlds meet

The job is the same as in any PWR: pass heat from the radioactive water to the clean water *without letting them mix*. The geometry is different again. Instead of a separate tank of U-tubes, this reactor uses **compact plate steam generators** — flat, finely channelled heat exchangers stacked in the annulus (six for power, plus two reserved for safety). Figure 3 opens one up.

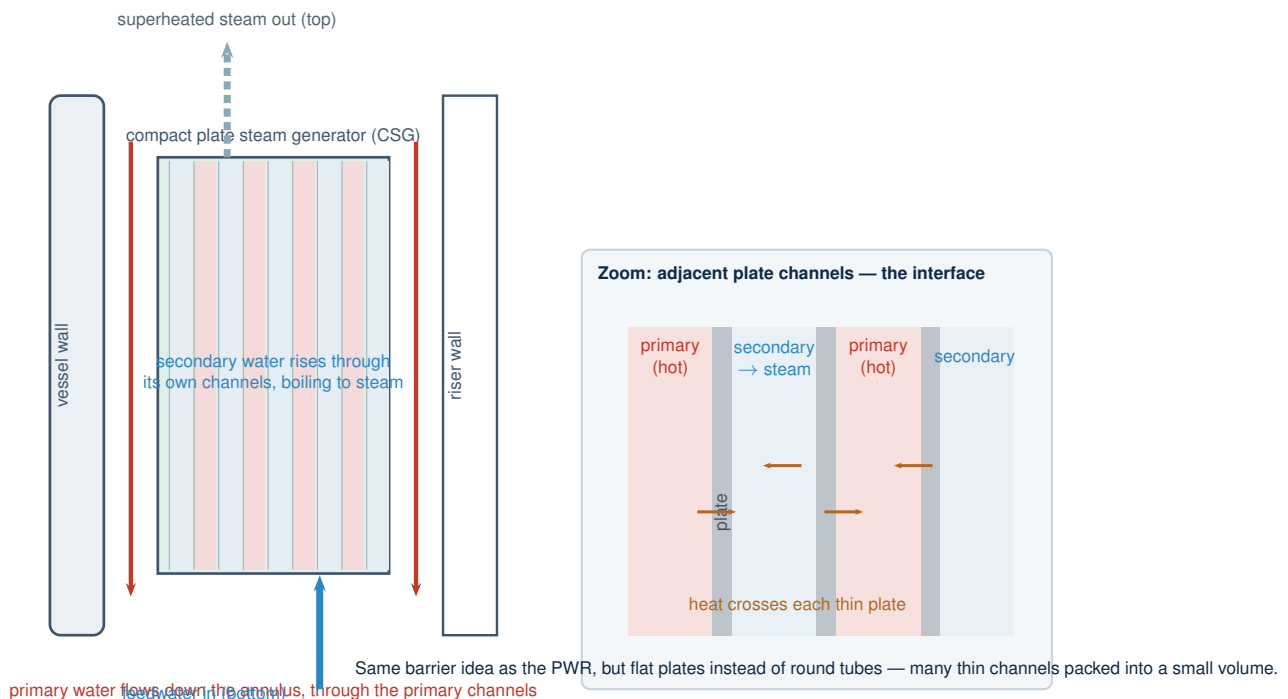


Figure 3: A compact plate steam generator, shown as one block in the annulus (left) with the key idea enlarged (right). Hot **primary** water flows down the annulus through its own channels; **feedwater** flows up through the *interleaved* secondary channels, boiling to **superheated steam** in a single pass (a “once-through” generator, so no steam drum or dryers are needed). The enlargement shows the barrier: primary and secondary run in adjacent channels separated by thin metal plates — the same heat-passes-atoms-don’t principle as the PWR’s U-tubes, but packed flat.

Why this barrier matters. As in the large plant, the plate wall keeps radioactivity penned in the primary water — and a slow, growing divergence on a steam-generator signal is exactly what the digital twin watches for. Folding the generators inside the vessel removes the large external pipes that, in a big PWR, are themselves something that can fail.

5 The fuel and the core

The fuel technology is the same as a PWR’s — ordinary uranium-dioxide fuel, no exotic materials — but there is less of it, and it is shorter. Figure 4 zooms from the core down to a single pellet.

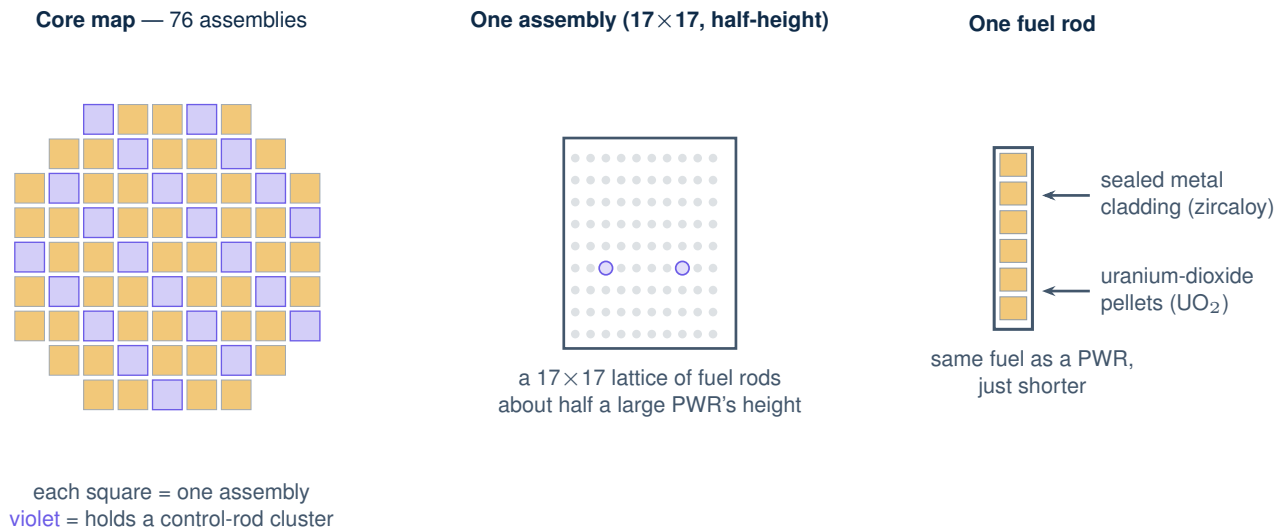


Figure 4: From core to pellet. The core holds **76 assemblies** — fewer than a large PWR's 150-plus — with some positions carrying control-rod clusters (violet). Each **assembly** is a standard 17×17 lattice of fuel rods, but only about *half the height*. Each **fuel rod** is the same sealed tube of **uranium-dioxide pellets**. Fewer, shorter assemblies mean lower total power and a longer time between refuellings.

5.1 The chain reaction, briefly

The physics is identical to the PWR's: uranium nuclei **fission** when they absorb neutrons, releasing heat and more neutrons, and the surrounding water both carries the heat away and *moderates* (slows) the neutrons so the reaction sustains. If the water heats or boils it moderates less, so power eases off — the negative feedback that makes any light-water reactor self-stabilising, and the same effect the demo's point-kinetics engine integrates live.

Running **boron-free** at power makes this feedback do more of the work: without dissolved boron to fine-tune reactivity, the design relies on a deliberately strong moderator response, so the small core settles itself. A smaller core also holds less total heat to remove after shutdown, which is part of why passive pool cooling can do the job.

6 Where each part lives in the console

As with the PWR, every part above has a home screen in the demo — and the SMR modules share the same fleet, plant view and tooling as the large units. The demo's Component Registry already models these modules as integral (integral reactor module, in-vessel steam generator, compact pumps, passive heat sink, compact containment), matching this guide.

Real component	Console screen
Core, fuel, chain reaction	<i>Reactor Kinetics</i> (the live physics), <i>Core</i> (parameters, core map)
Control rods (boron-free at power)	<i>Control Rods</i> (rod banks, SCRAM)
Integral vessel & in-vessel pumps	<i>Coolant / TH, 3D Reactor View</i> (renders as a smaller core)
Compact plate steam generators	<i>Steam & Secondary, Coolant / TH</i>
Integral pressuriser	<i>Coolant / TH</i>
Turbine, generator, condenser	<i>Steam & Secondary, Power & Grid</i>
Two units in one plant	<i>Plant Fleet, Plant 3D View</i> (modules feed a shared header)
Passive cooling & heat sink	<i>Radiation / Safety, Component Registry</i>
Measured-vs-modelled signals	<i>Digital Twin</i>

Note on the console. Selecting an SMR module switches the Coolant/TH and Steam & Secondary pages to this integral picture — in-vessel pumps, compact plate steam generators and an integral pressuriser, with no external loops — alongside the Component Registry, so the whole console reflects the NUWARD-type layout described here. (Selecting a large PWR unit shows its three- or four-loop arrangement instead.)